

Question	Answer	Marks													
10(a)	(decay is) not affected by external / environmental factors	B1													
10(b)(i)	half-life of X = $2T$ and half-life of Y = $3T$	B1													
	both samples show decay constant, in terms of $1 / T$, equal to $\ln 2 / \text{half-life}$ (decay constant of X = $\ln 2 / 2T$ and decay constant of Y = $\ln 2 / 3T$ if both half-lives correct)	B1													
	both samples show N_0 , in terms of AT , equal to initial activity / decay constant (N_0 for X = $8AT / \ln 2$ and N_0 for Y = $3AT / \ln 2$ if both decay constants correct)	B1													
	(Fully correct table: <table><tr><td></td><td>half-life</td><td>decay constant</td><td>A_0</td><td>N_0</td></tr><tr><td>X</td><td>$2T$</td><td>$\ln 2 / 2T$</td><td>$4A$</td><td>$8AT / \ln 2$</td></tr><tr><td>Y</td><td>$3T$</td><td>$\ln 2 / 3T$</td><td>A</td><td>$3AT / \ln 2$</td></tr></table>)			half-life	decay constant	A_0	N_0	X	$2T$	$\ln 2 / 2T$	$4A$	$8AT / \ln 2$	Y	$3T$	$\ln 2 / 3T$
	half-life	decay constant	A_0	N_0											
X	$2T$	$\ln 2 / 2T$	$4A$	$8AT / \ln 2$											
Y	$3T$	$\ln 2 / 3T$	A	$3AT / \ln 2$											
10(b)(ii)	correct substitution of A_0 and λ into $A_0 \exp (-\lambda t)$ for sample X or sample Y	C1													
	$4A \exp (-t \ln 2 / 2T) = A \exp (-t \ln 2 / 3T)$	C1													
	$t = 12T$	A1													
10(c)	<i>Any two points from:</i> (radiation) emitted in all directions, not just in direction of detector - some radiation absorbed by air / sample / window of detector - some radiation may not register even though it reaches detector	B2													